Ideation Phase

Brainstorm & Idea Prioritization

Date	13 July 2026
Project Name	India's Agricultural Crop Production Analysis using Tableau
Maximum Marks	4 Marks

SECTION 1 – PROBLEM IDENTIFICATION

Core Challenge:

⚠️ Large agricultural production datasets spanning multiple years (1997–2021) remain difficult to understand and analyse when stored in large spreadsheets.

⚠️ Key decision-makers cannot efficiently track long-term crop trends, state-wise performances, spatial distributions, or seasonal variations (Kharif, Rabi, Zaid).

⚠️ Calculating production metrics like agricultural yield relative to cultivated area changes requires slow, manual sorting routines that delay modern resource distribution.

SECTION 2 – IDEA GENERATION

Below is the conceptual structure of the structural alternative solutions generated during team ideation:

💡 **Idea 1** Excel Interactive Dashboard	💡 **Idea 2** Static PDF Planning Reports	💡 **Idea 3** Power BI Cloud Dashboard	💡 **Idea 4** Python Web Visualization Script
💡 **Idea 5 (Selected)** **Tableau Public Dashboard**	💡 **Idea 6 (Selected)** **Tableau Storyboard Presentation**	💡 **Idea 7** Geospatial Interactive Maps	💡 **Idea 8** Custom Web-based Analytics Tool

SECTION 3 – IDEA EVALUATION

Solution Idea	Ease of Use	Interactive Features	Scalability	Visual Quality	Complexity	Reporting
Excel Dashboard	High	Medium	Low	Low	Low	Medium
Static Reports	High	None	Low	Low	Low	Low
Power BI Dashboard	Medium	High	High	High	Medium	High

Python Visualization	Low	Medium	High	High	High	Medium
Tableau Dashboard + Story	**High**	**High**	**High**	**High**	**Low**	**High**
Interactive Maps	Medium	High	Medium	High	High	Medium
Web Analytics Engine	Medium	High	High	Medium	High	High

SECTION 4 – PRIORITIZATION MATRIX

💡 HIGH IMPACT – HIGH EFFORT

• Python Web Visualization Script
• Custom Web-based Analytics Tool
• Specialized Geospatial Interactive Maps

🏆 HIGH IMPACT – LOW EFFORT

✔ Tableau Public Dashboard & Story (Selected Solution)
• Power BI Cloud Dashboard Platform

💡 LOW IMPACT – HIGH EFFORT

• Manual cross-state spreadsheet systems

💡 LOW IMPACT – LOW EFFORT

• Static PDF Planning Reports
• Legacy Excel Dashboards

SECTION 5 – FINAL SELECTED IDEA

Why Tableau Public was Selected:

💡 **Interactive Dashboards & KPI Cards:** Provides real-time summaries of cultivated area changes and automated calculated yield metrics.

💡 **Advanced Filtering:** Allows immediate isolation of seasonal variables (Kharif, Rabi, Zaid) across individual states without manual coding.

💡 **Built-in Map Layers:** Features high-quality geospatial mapping to quickly spot dropping production zones.

💡 **Tableau Story Feature:** Combines independent interactive sheets into an ordered presentation flow that helps stakeholders understand agricultural trends and make informed, data-driven decisions.

💡 **Efficient Analytics Workflow:** Provides fast and interactive data exploration with minimal development effort, allowing users to analyse agricultural production efficiently.

SECTION 6 – CONCLUSION

Following a thorough evaluation of alternative tracking platforms, the integrated deployment of a **Tableau Public Dashboard and Tableau Story** provides the optimal visualization platform. It delivers the ideal balance of fast usability, superior visual representation quality, robust cross-season scalability, and native geospatial intelligence. This chosen path guarantees an intuitive, responsive analytics environment perfectly customized for evaluating historical trends across India's agricultural crop sectors.